

KINEMATIC MODELING OF SOFT CONTINUUM MANIPULATORS: A COMPARATIVE STUDY OF PIECEWISE CONSTANT CURVATURE AND COSSERAT ROD THEORY

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ABSTRACT

This paper integrates Piecewise Constant Curvature (PCC) and Cosserat Rod kinematic models into a Semantic Digital Twin framework. Using a symbolic math engine, we constructed multi-section soft robots with heterogeneous properties. We evaluated the models based on computational speed and workspace reachability. Evaluation shows that while PCC offers superior computational speed, the Cosserat model provides significantly greater workspace expressiveness by supporting axial torsion and longitudinal extension. This framework enables a trade-off between simulation performance and physical accuracy.

1. INTRODUCTION

Soft robotics are systems made from compliant materials that safely deform and adapt to their environment [1]. By utilizing flexible structures and various actuation methods such as fluidic pressure or smart materials [2], soft robots reach configurations difficult for traditional rigid links [1]. Their inherent dexterity and compliance enable operation in delicate spaces and safe human interaction [1, 2]. However, modeling these systems remains a challenge due to their continuous deformation and highly nonlinear behaviors [3, 4]. External factors like gravity or environmental contact further alter the robot's shape and kinematics, rendering traditional rigid-body modeling techniques insufficient [5, 3].

Simulation environments are essential for analyzing kinematics prior to robot deployment [6]. However, standard simulators are optimized for rigid links, necessitating model simplifications for real-time tasks [3]. Modeling approaches are generally categorized into geometric, continuum, and data-driven methods [7]. Analytical frameworks are often preferred over data-driven methods to avoid prediction uncertainty and provide general mappings applicable across various designs [4, 8]. The Piecewise Constant Curvature (PCC) model is a fundamental geometric approximation that simplifies the robot into circular arcs [8], offering the computational speed required for real-time monitoring [9].

Despite its efficiency, PCC has limitations in tasks involving complex spatial movements or external forces [7]. Geometric approximations often struggle to capture internal twisting (torsion) or the radial expansion seen in certain materials

under pressure [7, 10]. To achieve higher physical fidelity, Cosserat Rod theory provides a continuum-based differential approach. Instead of assuming a circular shape, it solves the robot's shape by integrating local strain rates along its length. This allows the model to represent 3D motion more accurately, capturing twisting and longitudinal stretching that geometric models ignore [7, 10].

In this work, we implement both the PCC and Cosserat Rod models within a simulation environment based on the Cognitive Robot Abstract Machine (CRAM) architecture [11]. To maintain computational efficiency, we utilize symbolic math wrappers that enable automatic differentiation. The primary contributions of this work are:

1. We develop a modular symbolic modeling framework for constructing multi-section continuum manipulators with heterogeneous physical properties, such as varying section lengths and radii.
2. We implement symbolic kinematic solvers for both Piecewise Constant Curvature and Cosserat Rod Theory, employing a 4th-order Runge-Kutta scheme to solve the differential rod equations within a semantic simulation environment.
3. We provide a comparative analysis evaluating the trade-offs between these two models, specifically focusing on computational latency and workspace expressiveness regarding axial torsion and longitudinal extension.

2. KINEMATIC MODELING

Kinematic modeling for soft continuum robots can be divided into two distinct sub-mappings [8]:

- The *actuation-to-configuration* mapping is robot-specific, converting hardware-dependent inputs such as pneumatic pressure or tendon tension into a mathematical description of the robot's shape.
- The *configuration-to-task-space* mapping is a general geometric mapping that relates this shape to the Cartesian pose of the robot in the task space.

This work focuses on the configuration-to-task-space mapping. By defining the degrees of freedom in the configuration space (e.g., curvature and bending plane), the implementation remains hardware-agnostic. This ensures that the simulation framework can be integrated with various soft robotic designs by providing the specific actuation-to-configuration mapping required for the hardware.

To represent this geometric relationship, we implement two distinct analytical approaches: the geometric Piecewise Constant Curvature (PCC) model and the differential Cosserat Rod model.

2.1. Piecewise Constant Curvature (PCC)

The PCC model is a geometric approximation that assumes the robot consists of one or more sections, each deforming into a perfect circular arc [8]. This assumption simplifies the infinite degrees of freedom into two parameters per section.

To describe the shape of a flexible rod of length L , we define:

1. **Curvature (κ):** The tightness of the bend (the inverse of the circle's radius).
2. **Bending Plane (ϕ):** The angle that determines which direction the robot leans relative to its base.

The total bending of a segment with arc length L is captured by the angle $\theta = \kappa L$. When $\kappa \neq 0$, the translation vector $\mathbf{p} = [p_x, p_y, p_z]^T$ representing the tip position relative to the base frame is defined by the following components:

$$p_x = \frac{\cos(\phi)(1 - \cos(\theta))}{\kappa} \quad (1)$$

$$p_y = \frac{\sin(\phi)(1 - \cos(\theta))}{\kappa} \quad (2)$$

$$p_z = \frac{\sin(\theta)}{\kappa} \quad (3)$$

The orientation of the tip frame is represented by a rotation matrix $R \in SO(3)$. This is computed by first rotating the base frame by ϕ to align the y -axis with the plane of bending, performing a rotation of θ around the new y -axis, and finally applying a counter-rotation of $-\phi$ to maintain local frame alignment:

$$R = R_z(\phi)R_y(\theta)R_z(-\phi) \quad (4)$$

The resulting homogeneous transformation matrix $T = \begin{bmatrix} R & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$ defines the pose of the tip. In our implementation, a singularity check is performed when the robot is perfectly straight ($\kappa \rightarrow 0$). In this limit, the equations simplify to a pure translation along the z -axis: $p_x = 0$, $p_y = 0$, and $p_z = L$, with R becoming the identity matrix.

2.2. Cosserat Rod Theory

PCC cannot model non-circular shapes or internal twisting. To overcome this, we implement Cosserat Rod Theory, which treats the robot as a continuous elastic string that can bend, twist, and stretch at every point [7].

Instead of assuming a fixed shape, Cosserat theory describes the rates of change along the robot's length. We represent the robot's configuration as a frame $T(s)$ that travels along the rod from the base ($s = 0$) to the tip ($s = L$). At any point s , the strain vector $\xi(s)$ tells the frame how to turn and move to reach the next point.

The change in the frame's pose is given by the following differential equation:

$$\frac{dT(s)}{ds} = T(s)\hat{\xi}(s) \quad (5)$$

where $\xi(s) = [u_x, u_y, u_z, v_x, v_y, v_z]^T$ is the strain vector. In this vector, (u_x, u_y) represent bending rates, u_z represents axial torsion (twist), and v_z represents longitudinal extension (stretch). We assume zero shear ($v_x = v_y = 0$) for thin manipulators. The $(\hat{\cdot})$ operator denotes the hat mapping from $\mathbb{R}^6$ to the Lie Algebra $se(3)$:

$$\hat{\xi} = \begin{bmatrix} 0 & -u_z & u_y & v_x \\ u_z & 0 & -u_x & v_y \\ -u_y & u_x & 0 & v_z \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (6)$$

Because this is a differential equation, there is no single formula to find the tip position. Instead, we perform numerical integration. We divide the rod into N small steps and use a 4th-order Runge-Kutta (RK4) method to accumulate the total transformation. This allows our simulation to represent complex 3D shapes, such as a rod that is twisted at the base but straight at the tip, or a manipulator that stretches under high pressure, which are behaviors that PCC is geometrically incapable of representing.

3. IMPLEMENTATION IN A SEMANTIC FRAMEWORK

The kinematic models are integrated into the CRAM simulation environment [11] using a symbolic math framework. By utilizing symbolic math, the robot's motion is defined as a computational graph. This allows the system to evaluate motion formulas efficiently and calculate the exact derivatives needed to reach a target, enabling real-time performance. To represent continuous bending within a standard rigid-body simulator, the soft structure is discretized into a serial chain of small rigid segments. These segments are linked to the underlying mathematical model, creating a smooth visual approximation. Finally, the robot is semantically registered as a functional *Manipulator* rather than a passive mesh. This metadata allows the framework to reason about the soft arm

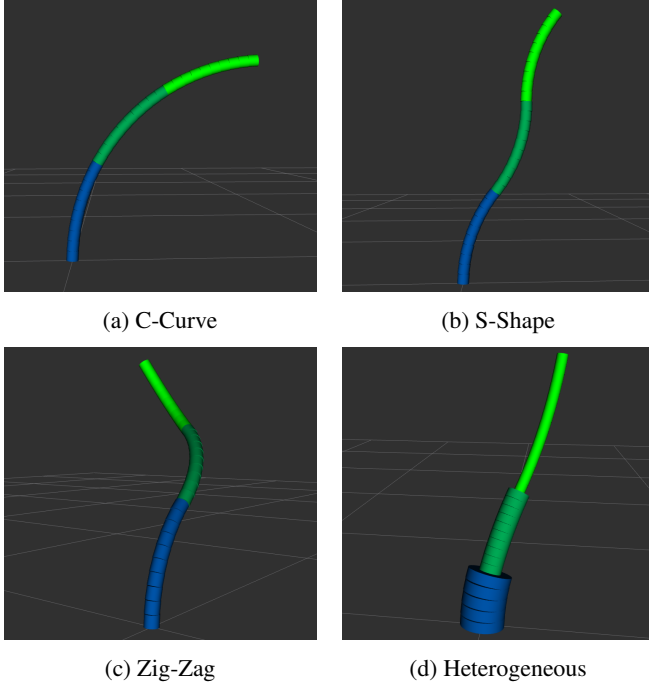


Fig. 1: Multi-section continuum robot configurations: (a) C-curve, three sections bending in the same direction; (b) S-shape, three sections bending in opposite directions (same plane); (c) Zig-Zag, three sections bending in different planes; (d) A heterogeneous robot featuring sections with varying lengths and radii.

as a tool capable of goal-directed interaction and precise manipulation in 3D space.

4. EVALUATION AND RESULTS

We evaluated the framework through qualitative shape generation tests and quantitative reachability benchmarks. All experiments were performed on a 1.0m soft manipulator.

4.1. Multi-Section Shape Generation

By connecting multiple sections with independent parameters, the system can generate complex spatial configurations. As shown in our simulations, the robot can successfully transition between simple C curves, S shapes, and Zig-Zag patterns (see figs. 1a to 1c). Our implementation also supports heterogeneous sections, allowing for varying lengths, radii, and visual resolutions within a single robot (see Fig. 1d).

The visual discretization proved effective. Using 10 segments per section provided a smooth visual representation while maintaining a stable frame rate in the simulation environment.

4.2. Torsion and Extension

To highlight the difference between geometric and differential models, we compared the PCC and Cosserat implementations under torsion and extension tasks.

- **Torsion:** As shown in Fig. 2, while PCC can bend in 3D space by rotating the bending plane, it cannot represent internal twisting around the robot’s spine. The Cosserat model allows the tip to rotate independently of its position.
- **Extension:** The PCC model assumes a constant arc length. Stretching in PCC would require external length updates. In contrast, the Cosserat model treats extension as an internal strain variable, allowing the structure to elongate or compress smoothly.

4.3. Computational Complexity and Speed

We benchmarked the average time required to solve a reaching task for both models. The results showed that the PCC model achieved an average solve time of 35.95 ms (± 3.44), while the Cosserat model required 538.71 ms (± 83.14). The Cosserat model is slower because it must solve for the robot’s shape through step-by-step numerical integration, whereas PCC uses a single direct formula.

The benchmark was conducted over 50 successful trials. The target positions were generated within a 3D bounding box relative to the robot’s base, defined by a horizontal range of ± 0.2 meters (X, Y) and a vertical height between 0.7 and 0.9 meters (Z). This ensures the targets were within the physical reach of both models. The solvers were configured with an integration step of $dt = 0.1$ and a maximum of 200 iterations.

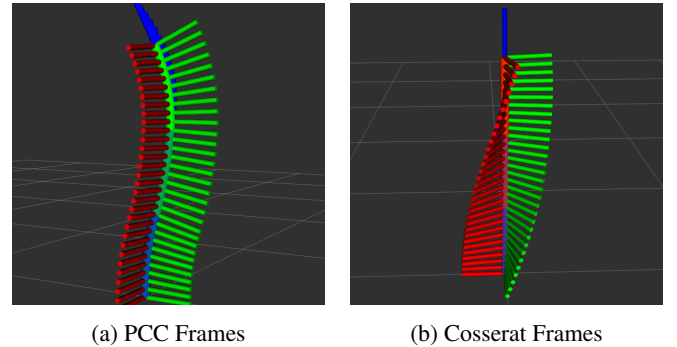
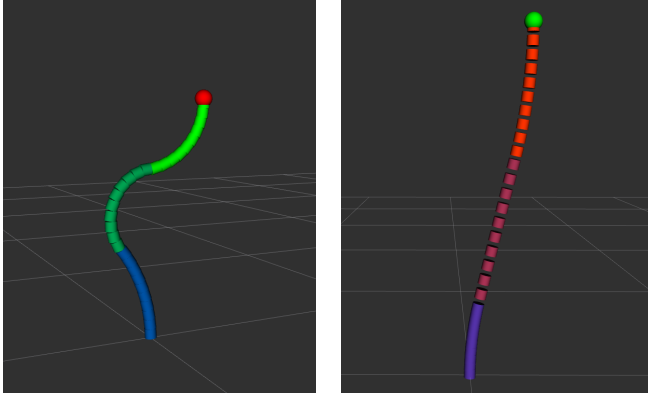


Fig. 2: Side-by-side of PCC vs Cosserat twisting, with the coordinate frames (TF tree) visible to show orientation of the tip. (a) PCC tip orientation is locked to the arc; (b) Cosserat tip is rotated 90°.



(a) PCC reaching target

(b) Cosserat reaching target

Fig. 3: Inverse Kinematics reaching tasks. (a) PCC model reaching a target (red sphere) within its length by curving. (b) Cosserat model reaching a distant target (green sphere) by curving and stretching.

4.4. Workspace Reachability

To evaluate the capabilities of both models, we conducted three experiments across varying target volumes. We distinguished between position-only tasks and full-pose tasks requiring a specific orientation ($\text{yaw} \in [-\pi, \pi]$). The results are summarized in Table 1.

4.4.1. Experiment 1: Proximal Reach

Targets were generated within a small volume reachable by a 1.0m arc ($Z \in [0.7, 0.9]$). In the position-only task, both models achieved a 100% success rate. As shown in Fig. 3a, the PCC model successfully converges to these proximal targets by solving for the required curvature and bending plane. However, when a yaw constraint was introduced, the PCC success rate plummeted to 15%, while the Cosserat model maintained 100%. This is because in PCC, the tip cannot rotate around its own axis without displacing its position. In contrast, the Cosserat model utilizes its independent torsion degree of freedom to match the target orientation perfectly.

Table 1: Inverse Kinematics Success Rates (%) across Target Volumes

Target Volume (Range Z in meters)	PCC Model		Cosserat Model	
	Pos.	Pose (+Yaw)	Pos.	Pose (+Yaw)
Exp 1: $Z \in [0.7, 0.9]$	100.0	15.0	100.0	100.0
Exp 2: $Z \in [0.2, 1.3]$	51.0	21.0	100.0	100.0
Exp 3: $Z \in [0.05, 3.0]$	4.0	2.0	81.0	84.0

4.4.2. Experiment 2: Distal Reach

Targets were sampled in a volume extending beyond the robot’s physical rest length ($Z \in [0.2, 1.3]$). The PCC model reached only 51% of targets in the position-only task, failing all points where the required arc length exceeded 1.0m. The Cosserat model achieved a 100% success rate in both configurations. As illustrated in Fig. 3b, the Cosserat model reaches these distal targets by optimizing its longitudinal extension strain (v_z).

4.4.3. Experiment 3: Boundary Stress Test

We tested the models against a large volume with targets up to 3.0m ($3 \times$ rest length). The PCC model effectively failed this test ($\leq 4\%$ success). Remarkably, the Cosserat model maintained a success rate of over 80%, reaching targets far beyond the initial material length.

5. CONCLUSION

This work presented a symbolic framework for integrating continuum kinematics into a Semantic Digital Twin. Our evaluation highlights a fundamental trade-off: the Piecewise Constant Curvature (PCC) model is nearly 15 times faster and ideal for high-speed visualization within a fixed workspace, but it is limited by orientation coupling and fixed arc lengths. In contrast, the Cosserat Rod model is necessary for tasks requiring longitudinal elongation or axial twisting, providing a significantly denser reachable workspace. This framework balances computational efficiency against physical fidelity depending on the specific manipulation task. Future work will focus on integrating external force dynamics to simulate environmental interactions and gravity-induced deformations.

6. REFERENCES

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